

Section C - Description/Specifications/Statement of Work

STATEMENT OF WORK

Microwave Electronics Systems and Microsystems Engineering Support for the Radar Division Advanced Concept Group

1.0 INTRODUCTION

This Statement of Work describes the scientific and engineering support to be provided to the Advanced Concepts Group of the Radar Division of the Naval Research Laboratory (NRL), Washington, DC. NRL seeks contractor technical assistance in the highly specialized areas of Radio Frequency (RF) components and systems, electronic and microelectronics systems design and subsystems integration, digital signal processing and embedded microprocessors, related signal processing hardware and software, antennas and electromagnetics, wireless systems testing and demonstrations, and other engineering specialties. Tasks will be performed at the contractor's facilities, NRL-DC, NRL Chesapeake Bay Detachment at Chesapeake Beach, MD, NRL Blossom Point Test Facility, and at other field sites. Some work may be performed aboard ship or aircraft.

1.1 Background

The Advanced Concepts Group plans, encourages and fosters scientific research of microwave and millimeter-wave technologies in support of future Navy power in the space, air, and maritime domains. This charter includes research and development activities on microwave power beaming transmitting and receiving systems and on advanced/exploratory electronics systems technologies leading to improvements in the size, weight, power, performance, cost, reliability, and maintainability of new and existing Navy radar, communications, and directed energy systems. Services provided under this SOW range from top-level concept development to detailed design, fabrication, test and operation of subsystems and components. A wide range of experience in radar and electronic warfare as well as a variety of scientific, engineering and support backgrounds will be necessary to be successful. The required services include numerous interrelated tasks that vary in both their technical dimensions and deliverable product.

1.2 Scope

The Contractor will provide research and development (R&D), radar and RF system analysis, systems engineering, and computer programming in support of the following basic task areas:

Task I – RF Systems Engineering

Task II – Digital Signal Processing and Embedded Processing

Task III – Antennas and Wireless Link Engineering

Task IV – Software Engineering and Development

Task V – Electronic Hardware Engineering and Development

Task VI – Mechanical/Aerospace Engineering

Task VII – Control Systems Engineering

Task VIII- RF Systems and Electromagnetic Modeling and Simulation

Task IX – Electromechanical Technician

Task X – Technical Writer

Task XI – Project Support Manager

Task XII – Wireless Systems Application Engineering

2.0 GENERAL REQUIREMENTS

The Contractor shall have its support team in place and fully functioning at the time of the base contract award or exercise of a contract option.

During this period the Contractor will execute the technical requirements of either the base or option of the contract in a manner that provides for high quality, timely support while incorporating the proper mix and the most effective use of personnel.

The Base Technical Requirements for the contract award is in Paragraph 3.0 Base Technical Requirements

The Option Technical Requirements for the contract award is in Paragraph 4.0 Technical Requirements Option

2.1 Non-Personal Services

The Government will neither supervise Contractor employees nor control the method by which the Contractor performs the required tasks. The Government will not assign tasks to, or prepare work schedules for, individual Contractor employees.

The Contractor shall be responsible for managing its employees and guarding against any actions that are of the nature of personal services, or give the perception of personal services as defined in FAR-Part 37, Service Contracting.

The Contractor shall notify the Contracting Officer (KO) if any Government-requested actions constitute, or are perceived to constitute, personal services.

2.2 Business Relations

The Contractor shall integrate and coordinate all activity needed to execute this contract.

2.3 Contract Administration and Management

The following subsections specify requirements for contract management and Contractor personnel administration.

2.3.1 Contract Management

The Contractor shall establish clear organizational lines of authority and responsibility to ensure effective management of the resources assigned to this contract.

2.3.2 Personnel Administration

The Contractor shall assign a single point of contact, also known as the Program Manager (PM), who must work closely with the Government Contracting Officer and Contracting Officer's Representative (COR), as applicable.

The Contractor PM must ultimately be responsible for ensuring that the Contractor's performance meets all government contracting requirements within cost and schedule. The PM must have the requisite authority for full control over all company resources necessary for contract performance. As appropriate, the role of the PM may be a full-time management position, or a collateral duty assigned to a technical team member.

The Contractor shall make necessary travel arrangements for Contractor employees only after the travel has been reviewed and approved by the COR.

2.3.3 Contract Administration

The Contractor shall establish processes and assign appropriate resources to effectively administer this contract.

The Contractor shall respond to Government requests for contractual actions within five (5) working days.

2.3.4 Contractor's Management Plan (CMP)

The Contractor shall submit a Contractor's Management Plan (CMP) defining the Contractor's approach to implementing the contract.

The CMP provides the Government insight into the systems, processes, and structure within which the Contractor operates. The CMP should include, as applicable and if not defined in a separate plan: quality control, risk management, systems engineering, software development, configuration management, earned value management, and subcontract management.

A draft CMP is to be delivered with the Contractor's proposal; three copies of the complete version shall be submitted 30 days after contract award to the KO and COR, and within 5 working days when changes are made thereafter.

The Contractor shall receive the Contracting Officer's acceptance in writing of any proposed change to the management plan once the CMP has been accepted by the Government.

3.0 INITIAL BASE TECHNICAL REQUIREMENTS

The following section specifies the base technical requirements of the contract. The Government will assign specific projects during the contract

performance via Technical Instructions tasked by the COR in accordance with the clause in Section H of the solicitation and resultant contract entitled “Technical Instructions”.

3.1 Project Support Manager

- 3.1.1 Manage Project design efforts to include schedule, funding and day to day project oversight;
- 3.1.2 Provide weekly project status with Principal Investigator (PI) and Contracting Officer’s Representative (COR);
- 3.1.3 Assist the PI and COR with acquisition planning, execution and administration;
- 3.1.4 Evaluate, validate, and manage project change requests;
- 3.1.5 Prepare and monitor project schedules and budgets; provide technical guidance and assistance on contracting and assisting with direct design efforts field work as needed;
- 3.1.6 Optimize performance, cost, and schedule by ensuring project objectives are achieved on schedule and at the lowest cost compatible with PI require monitor technical progress vs schedule and budget.
- 3.1.7 Provide regular technical and financial reporting to Sponsor and the PI;
- 3.1.8 Provide document and inventory control support.

3.2 RF Systems Engineering

The Contractor shall provide subject matter expertise in RF Systems Engineering. Technical instructions may be issued for the execution of projects that include, but are not limited to:

- 3.2.1 Perform system analysis, design, development, and laboratory and/or field-test of RF systems (radar, communications, and/or electronic warfare);
- 3.2.2 Conduct trade studies that lead to optimal systems performance for a defined set of conditions and applications;
- 3.2.3 Perform analysis, test, evaluation, and define improvements to existing (legacy) RF systems;
- 3.2.4 Investigate new RF component technologies and their application to Navy or DoD systems;
- 3.2.5 Perform laboratory measurements and tests of subsystems in support of Navy or DoD RF system development;
- 3.2.6 Script and conduct automated, parametric tests of RF and electronics subassemblies using tools such as Python, MATLAB instrument control toolbox, NI TestStand, or NI LabView.
- 3.2.7 Perform component and transceiver measurements of parameters such as S-parameters, P1dB, OIP3, noise figure, gain, leakage, isolation, etc.
- 3.2.8 Perform technical analysis and computer simulations to evaluate component and subsystem application to Navy or DoD systems;
- 3.2.9 Fully document test results and analysis;
- 3.2.10 Produce and deliver developed systems and subsystems;
- 3.2.11 Develop technical and user manuals for developed equipment;
- 3.2.12 Participate in technical meetings and design reviews, provide critiques, review documentation, and monitor technical progress.

3.3 Wireless Systems Application Engineering

The Contractor shall provide subject matter expertise in the development and demonstration of classified microwave and wireless systems for military applications. Technical instructions may be issued for the execution of projects that include, but are not limited to, the following, in addition to the requirements in the prior subsection:

- 3.3.1 Design and development and implementation of radar and/or electronic warfare (EW) systems for Navy or Marine Corps applications, including RF, millimeter wave, W band sensors, communications systems, radar systems, and microwave power beaming links.
- 3.3.2 Interaction with military personnel, defense contractors, and intelligence organizations in support of requirements development based on platform and threat vulnerabilities, experiment planning, and demonstration execution
- 3.3.3 Development of high reliability hardware and software for extreme environmental conditions, including space and maritime applications.
- 3.3.4 Handling, storing, and maintaining DoD classified project digital data, including data generated in experimental testing

4.0 TECHNICAL REQUIREMENTS

The following section specifies the technical requirements options for the contract. The Government will assign specific projects during the contract

performance via Technical Instructions tasked by the COR in accordance with the clause in Section H of the solicitation and resultant contract entitled “Technical Instructions”.

4.1 RF Systems Engineering

The Contractor shall provide subject matter expertise in RF Systems Engineering. Technical instructions may be issued for the execution of projects that include, but are not limited to:

- 4.1.1 Perform system analysis, design, development, and laboratory and/or field-test of RF systems (radar, communications, electronic warfare, and/or microwave power beaming);
- 4.1.2 Conduct trade studies that lead to optimal systems performance for a defined set of conditions and applications;
- 4.1.3 Perform analysis, test, evaluation, and define improvements to existing (legacy) RF systems;
- 4.1.4 Investigate new RF component technologies and their application to Navy or DoD systems;
- 4.1.5 Perform laboratory measurements and tests of subsystems in support of Navy or DoD RF system development;
- 4.1.6 Script and conduct automated, parametric tests of RF and electronics subassemblies using tools such as Python, MATLAB instrument control toolbox, NI TestStand, or NI LabView.
- 4.1.7 Perform component and transceiver measurements of parameters such as S-parameters, P1dB, OIP3, noise figure, gain, leakage, isolation, etc.
- 4.1.8 Perform technical analysis and computer simulations to evaluate component and subsystem application to Navy or DoD systems;
- 4.1.9 Fully document test results and analysis;
- 4.1.10 Produce and deliver developed systems and subsystems;
- 4.1.11 Develop technical and user manuals for developed equipment;
- 4.1.12 Participate in technical meetings and design reviews, provide critiques, review documentation, and monitor technical progress.

4.2 Wireless Systems Application Engineering

- 4.2.1 Design and development and implementation of radar and/or electronic warfare (EW) systems for Navy or Marine Corps applications, including RF, millimeter wave, W band sensors, communications systems, radar systems, and microwave power beaming links.
- 4.2.2 Development of high reliability hardware and software for extreme environmental conditions, including space and maritime applications.
- 4.2.3 Interaction with military personnel, defense contractors, and intelligence organizations in support of requirements development based on platform and threat vulnerabilities, experiment planning, and demonstration execution
- 4.2.4 Handling, storing, and maintaining DoD classified project digital data, including data generated in experimental testing

4.3 Digital Signal Processing and Embedded Processing

The Contractor shall provide subject matter expertise in signal processing. Technical instructions may be issued for the execution of projects that include but are not limited to:

- 4.3.1 Develop embedded signal processing systems using general purpose processors (GPPs), field programmable gate arrays (FPGAs), microcontrollers, graphics processing units (GPUs);
- 4.3.2 Formulate, develop, and demonstrate new digital signal processing concepts and algorithms for spaceborne, shipboard, airborne, and land-based RF systems;
- 4.3.3 Apply new signal processing technologies to improve performance and/or provide new capabilities to RF systems;
- 4.3.4 Develop real-time image processing algorithms and human interfaces;
- 4.3.5 Analyze measured and simulated data;
- 4.3.6 Perform laboratory and field tests of experimental and operational systems;
- 4.3.7 Document test results and analysis;
- 4.3.8 Participate in technical meetings and design reviews, provide critiques, review documentation and monitor technical progress.

4.4 Antenna and Wireless Link Engineering

The Contractor shall provide subject matter expertise in electromagnetics engineering. Technical instructions may be issued for the execution of projects that include but are not limited to:

- 4.4.1 Conduct basic and applied research and development in electromagnetic (EM) theory and applications;

- 4.4.2 Conduct research in antenna design;
- 4.4.3 Develop and test advanced antenna and phased array systems;
- 4.4.4 Identify and develop improvements to deployed antenna systems;
- 4.4.5 Perform antenna and radar cross section measurements in a compact range facility;
- 4.4.6 Perform theoretical analysis on the transmission, propagation, and scattering of EM waves;
- 4.4.7 Analyze the radar signature of various ship classes using both calculated and measured data;
- 4.4.8 Develop new, or improve existing, techniques for calculating the radar scattering from ships and other vehicles;
- 4.4.9 Rigorously calibrate transmit and received power in wireless link experimentation;
- 4.4.10 Develop antenna measurement test plans and test reports;
- 4.4.11 Document test results and analysis;
- 4.4.12 Participate in technical meetings and design reviews, provide critiques, review documentation and monitor technical progress.

4.5 Software Engineering and Development

The Contractor shall provide subject matter expertise in software engineering and software development. Technical instructions may be issued for the execution of projects that include but are not limited to:

- 4.5.1 Define, design, and develop software to perform real-time control of RF and data acquisition systems;
- 4.5.2 Develop software to perform real-time signal and data processing required for RF systems;
- 4.5.3 Develop software that manages and directs multiple RF systems;
- 4.5.4 Develop software to model and simulate RF systems and their environment;
- 4.5.5 Develop, maintain, and enhance complex and diverse software systems based upon documented requirements, including real-time and near-real time performance;
- 4.5.6 Develop graphical user interfaces;
- 4.5.7 Provide specific input to the software components of a system design to include hardware/software trade-offs, software reuse, use of COTS/GOTS in place of new development, and requirements analysis and synthesis from system level to individual software components
- 4.5.8 Develop software to control laboratory equipment;
- 4.5.9 Develop high-level software architectures;
- 4.5.10 Thoroughly document developed software;
- 4.5.11 Participate in technical meetings and design reviews, provide critiques, review documentation and monitor technical progress.

4.6 Radiofrequency Electronic Hardware Engineering and Development

The Contractor shall provide subject matter expertise in hardware engineering and development. Technical instructions may be issued for the execution of projects that include but are not limited to:

- 4.6.1 Design, develop, and test analog/RF/microwave circuits and subsystems. This includes the design, fabrication, and testing of RF and microwave up- and down-converters, multilayer RF printed circuit boards,
- 4.6.2 Design, develop, and test digital circuits and subsystems, including the design, fabrication and testing of circuits to perform digital functions such as up- down-conversion, filtering, waveform generation, digitization, and beamforming;
- 4.6.3 Design, develop, and operate test equipment suites for the evaluation and verification of subsystems and systems;
- 4.6.4 Execute linear and nonlinear RF component measurements and analysis and/or wireless systems parameter testing
- 4.6.5 Operate, and program as appropriate, test equipment;
- 4.6.6 Design and develop integrated microwave modules and assemblies, including transmit/receive (T/R) modules. Designs must address mechanical, thermal, and cost considerations and constraints. Apply test results, including linear and nonlinear RF component measurements, to analysis and/or wireless systems parameter testing;
- 4.6.7 Produce and deliver developed systems and subsystems;
- 4.6.8 Document designs and test results;
- 4.6.9 Develop technical and user manuals for developed equipment;
- 4.6.10 Participate in technical meetings and design reviews, provide critiques, review documentation and monitor technical progress.

4.7 Mechanical/Aerospace Engineering

The Contractor shall provide subject matter expertise in mechanical and aerospace engineering and equipment development. Technical instructions may be issued for the execution of projects that include but are not limited to:

- 4.7.1 Design, fabricate, and test airborne structures, equipment, and electromechanical pointing systems;
- 4.7.2 Integrate and install equipment and structures with air frames;
- 4.7.3 Lead in the flight qualification of equipment and structures;
- 4.7.4 Design and develop thermal management and environmental control in airborne or spaceborne platforms;
- 4.7.5 Develop CAD models in Solidworks;
- 4.7.6 Conduct thermal analysis of simple cooling systems
- 4.7.7 Conducting structural analysis of bolted joints and mechanical structures and identify failure modes
- 4.7.8 Develop vibration and shock isolation for spacecraft, aircraft and shipboard applications
- 4.7.9 Perform finite element analysis for structural and thermal applications
- 4.7.10 Manage mass properties (e.g. mass distribution, center of gravity, etc.) and analyze thermal, vibration, load, and stress using CAD modeling
- 4.7.11 Lead in the development and integration of inertial navigation systems and integration with automatic flight control systems;
- 4.7.12 Fully document designs, and test results;
- 4.7.13 Participate in technical meetings and design reviews, provide critiques, review documentation and monitor technical progress.

4.8 Control Systems Engineering

The Contractor shall provide subject matter expertise in control systems engineering and equipment development. Technical instructions may be issued for the execution of projects that include but are not limited to:

- 4.8.1 Design, fabricate, and test electromechanical pointing systems;
- 4.8.2 Integrate and install equipment;
- 4.8.3 Analyze control system performance;
- 4.8.4 Optimize control loop stability and performance;
- 4.8.5 Ensure high performance shielding and grounding principles and practices are followed;
- 4.8.6 Program embedded and distributed control systems;
- 4.8.7 Fully document designs, and test results;
- 4.8.8 Participate in technical meetings and design reviews, provide critiques, review documentation and monitor technical progress.

4.9 RF Systems and Electromagnetic Modeling and Simulation

The Contractor shall provide subject matter expertise in modeling and simulation. Technical instructions may be issued for the execution of projects that include but are not limited to:

- 4.9.1 Design and develop models and simulations of radar systems;
- 4.9.2 Design and develop models of combat and weapon systems;
- 4.9.3 Design, develop, or modify propagation models;
- 4.9.4 Perform detailed analysis and simulations using M&S tools;
- 4.9.5 Use and modify established modeling tools;
- 4.9.6 Fully document designs, and test results;
- 4.9.7 Participate in technical meetings and design reviews, provide critiques, review documentation and monitor technical progress.

4.10 Technician

The Contractor shall provide laboratory technician support. The Government may issue Technical instructions for the execution of projects that include but are not limited to:

- 4.10.1 Manage, maintain, and operate a Compact Range Facility;

- 4.10.2 Manage, maintain and operate a machine shop;
- 4.10.3 Assemble antennas, arrays, RF hardware, and associated test fixtures;
- 4.10.4 Operate RF test equipment, test fixtures, and data acquisition systems;
- 4.10.5 Generate multilayer printed wiring board layouts based on engineering schematics and vendor-supplied design-for-manufacturing guidance.
- 4.10.6 Assemble and/or rework complex printed wiring board assemblies, including QFN packages, BGAs, and SMT “0201” components.
- 4.10.7 Perform basic verification and troubleshooting of electronic assemblies.
- 4.10.8 Manage the life cycle process for laboratory test and measurement equipment;
- 4.10.9 Maintain inventory control of laboratory equipment and consumable stocks;
- 4.10.10 Assist in the procurement and disposal of laboratory equipment and consumable stocks;
- 4.10.11 Generate test plans;
- 4.10.12 Provide logistical support for experiments and field tests;
- 4.10.13 Document and report the results of test, experiments or assessments;

4.11 Technical Writer

The Contractor shall provide subject matter expertise in Technical Writing. Technical writing instructions may be issued for the execution of projects that include, but are not limited to:

- 4.11.1 Edit and Proofread government reports and analyses
- 4.11.2 Research and documenting findings in various subject areas using a variety of media
- 4.11.3 Combining information from several authors into one cohesive document
- 4.11.4 Researching, writing and editing formal research papers, product descriptions and users manuals

4.12 Project Support Manager

The Contractor shall provide subject matter expertise in Project Support Management. Instructions may be issued for the execution of projects that include, but are not limited to:

- 4.12.1 Manage Project design efforts to include scheduling, funding and day to day project oversight
- 4.12.2 Provide weekly project status with Principle Investigator (PI) and Contracting Officer’s Representative
- 4.12.3 Assist the PI and COR with acquisition planning, execution and administration
- 4.12.4 Evaluate, validate, and manage project change requests
- 4.12.5 Prepare and monitor project schedules and budgets; provide technical guidance and assistance on contracting and assisting with direct design effort field work as needed.
- 4.12.6 Optimize performance, cost, and scheduling by ensuring project objectives are achieved on schedule and at the lowest cost compatible with PI requirements; monitor technical progress vs schedule and budget
- 4.12.7 Provide regular technical and financial reporting to Sponsor and the PI
- 4.12.8 Provide document and inventory control support

4.13 Wireless Systems Application Engineering

The Contractor shall provide subject matter expertise in Wireless Systems Application Engineering. Instructions may be issued for the execution of projects that include, but are not limited to:

- 4.13.1 Design and development and implementation of radar and/or electronic warfare (EW) systems for Navy or Marine Corps applications, including RF, millimeter wave, W band sensors, communications systems, radar systems, and microwave power beaming links.
- 4.13.2 Develop of high reliability hardware and software for extreme environmental conditions, including space and maritime applications.
- 4.13.3 Interact with military personnel, defense contractors, and intelligence organizations in support of requirements development based on platform and threat vulnerabilities, experiment planning, and demonstration execution
- 4.13.4 Handling, storing, and maintaining DoD classified project digital data, including data generated in experimental testing
- 4.13.5 Execute systems demonstrations for Warfighters and VIPs under challenging conditions and significant schedule pressure.

5.0 DELIVERABLES

5.1 Monthly Labor and Financial Status Report (Exhibit A)

- 5.1.1 Funding history, showing when and how much money was added to each ACRN for each person on the task, amounts allocated to labor, travel, and material, along with cumulative task funding for labor, travel, and material;
- 5.1.2 Funding balance for labor, travel, and materials in each ACRN and by person at the beginning of the monthly reporting period;
- 5.1.3 Labor hours and amounts expended for labor, travel, and materials in each ACRN and by person during the monthly reporting period;
- 5.1.4 Funding balance for labor, travel, and materials in each ACRN and by person at the end of the monthly reporting period.

5.2 Task Technical Progress Reports (Exhibit A)

The Contractor shall provide a monthly summary report of progress on each task of the contract that includes the following information:

- 5.2.1 Description of the work accomplished;
- 5.2.2 Summary of meetings attended;
- 5.2.3 Plans for next reporting period;
- 5.2.4 Summary of issues or concerns, if any.

5.3 Meeting Reports (Exhibit A)

The Contractor shall provide a summary of all meetings attended on behalf of the Radar Division. The report shall include discussion of the purpose of the meeting, attendees, information discussed, and the outcome/results of the meetings. If any actions are assigned to NRL, those are to be fully described.

5.4 Technical Reports & Presentation Materials (Exhibit A)

The Contractor shall provide technical reports and presentations as required.

5.5 Contractor's Management Plan (CMP) (Exhibit A)

The Contractor shall deliver a CMP under CDRL A003 containing the required information specified in SOW paragraph 2.3.4.

6.0 SECURITY REQUIREMENTS

The following are required at the time of award:

All Contractor personnel who require NRL base access or access to NRL unclassified material and/or unclassified systems shall possess a favorably completed DoD Tier 3 investigation.

Contractors who require access to Secret material must have, at a minimum, a favorably completed DoD Tier 3 investigation and final DoD-granted Secret security clearance.

Contractors who require access to Top Secret material must have, at a minimum, a favorably completed DoD Tier 5 investigation and final DoD-granted Secret security clearance.

The Contractor shall ensure that all classified material is handled in accordance with the issued DD 254, the National Security Program Operating Manual (NISPOM) (DoD 5220.22M) and all NRL and applicable Security Program Guides/Directives.

All Contractors (including subcontractors) identified in the Statement of Work shall supplement their current security practices by requiring any personnel involved in executing the contract to complete Government-sponsored and administered Operations Security (OPSEC) training, OPSE-1301 and any OPSEC guidance that may pertain to the project.

All Contractors (including subcontractors) issued a DoD interim SECRET security clearance may ONLY work on unclassified tasks in accordance with this contract SOW. No classified access will be authorized. Unclassified IA or IT positions are NOT authorized with a DoD issued interim SECRET clearance.